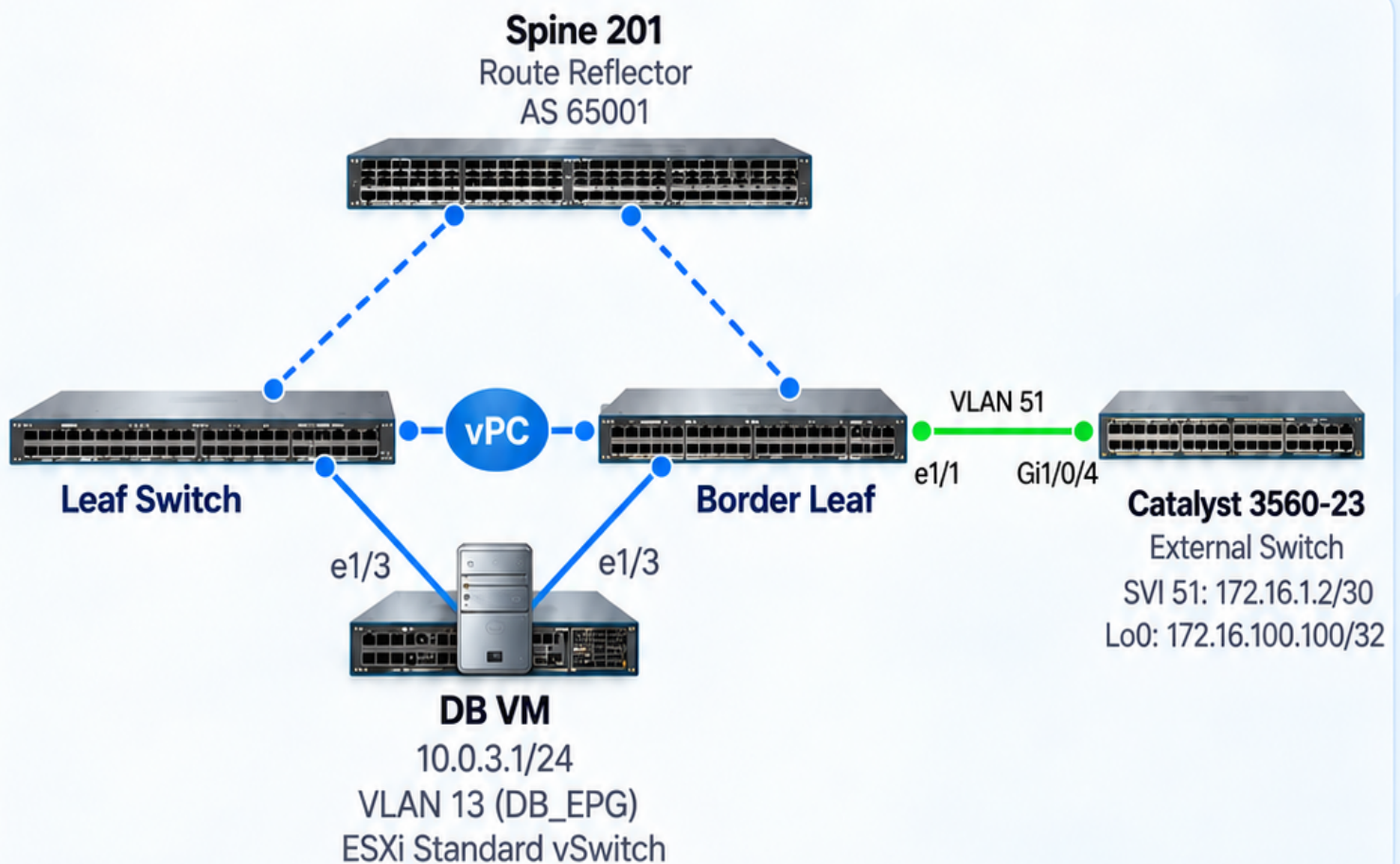


Cisco ACI

External Layer 3

Network Connectivity

Connecting the ACI fabric to external networks using OSPF routing



Author

Titus Majeza



<https://github.com/TitusM/Cisco-Data-Center>

External Layer 3 (L3Out) Connection

Introduction

Cisco Application Centric Infrastructure (ACI) allows you to connect the fabric to outside networks using standard routing protocols that operate on Layer 3. Such connections are referred to as Layer 3 outside (L3Out). They define the interfaces, protocols, and protocol parameters that are used to provide IP connectivity to external devices through routing protocols or static routes. A border leaf is a Cisco ACI leaf that provides Layer 3 connections to outside networks.

Cisco ACI supports the following routing mechanisms: static routes, Open Shortest Path First version 2 (OSPFv2), Enhanced Interior Gateway Routing Protocol (EIGRP), and Border Gateway Protocol (BGP). Policy enforcement through contracts is required with external networks just like a normal endpoint group (EPG) to EPG communications. However, the classification of external IP addresses/prefixes to an EPG (called external EPG or L3Out EPG) differs from a normal EPG.

This lab shows how to deploy OSPF routing between an ACI leaf and an adjacent Cisco Catalyst switch, and enable communication of an internal EPG (DB_EPG) endpoint with the external OSPF network.

Physical & Logical Topology

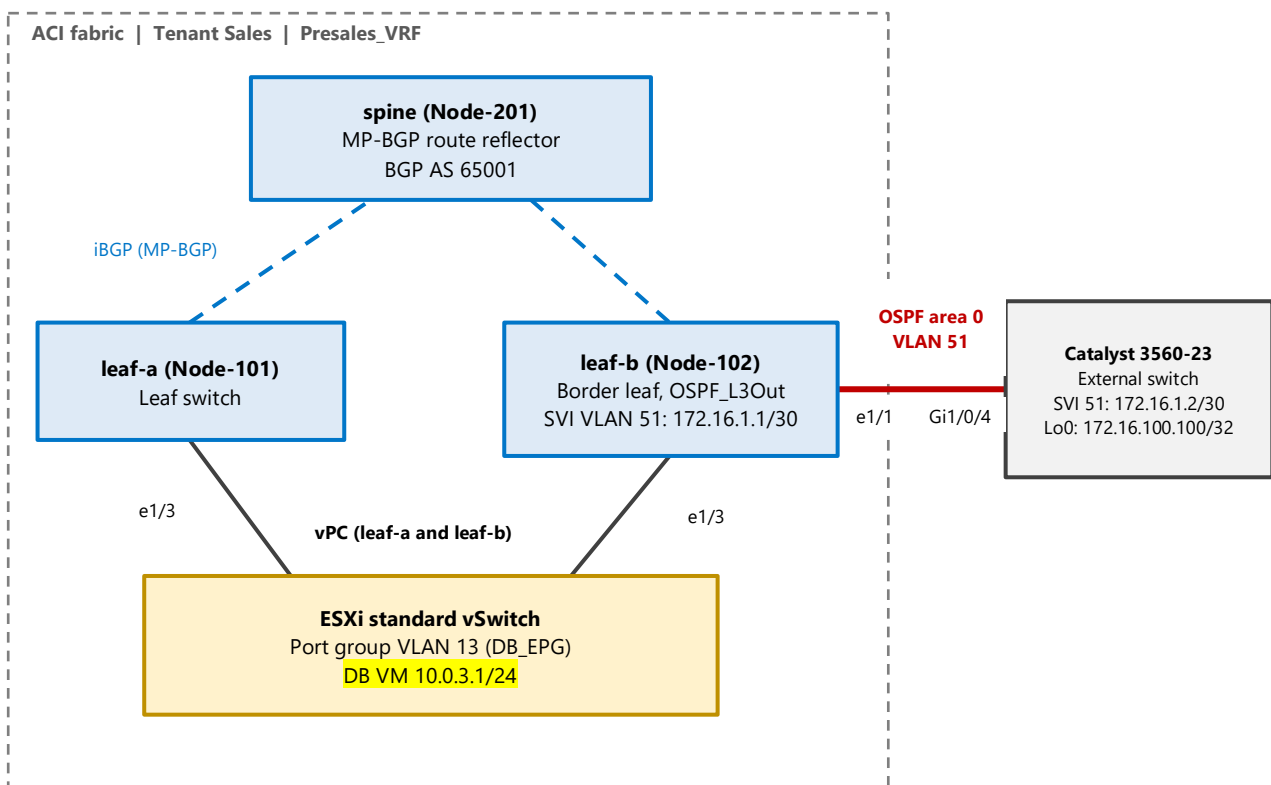
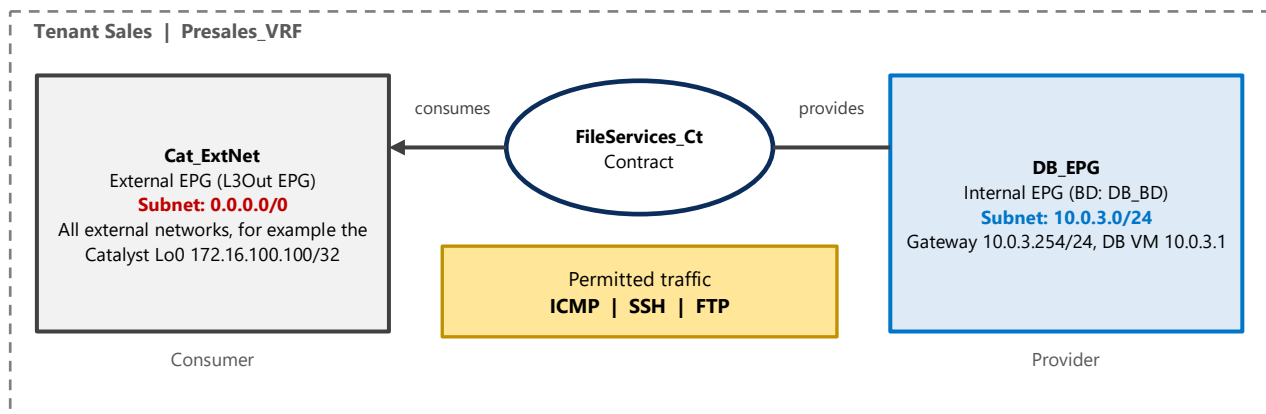


Figure: L3Out scenario. OSPF area 0 between leaf-b and the Catalyst switch over VLAN 51, MP-BGP route reflection through spine 201, and the DB VM (10.0.3.1) behind the vPC.

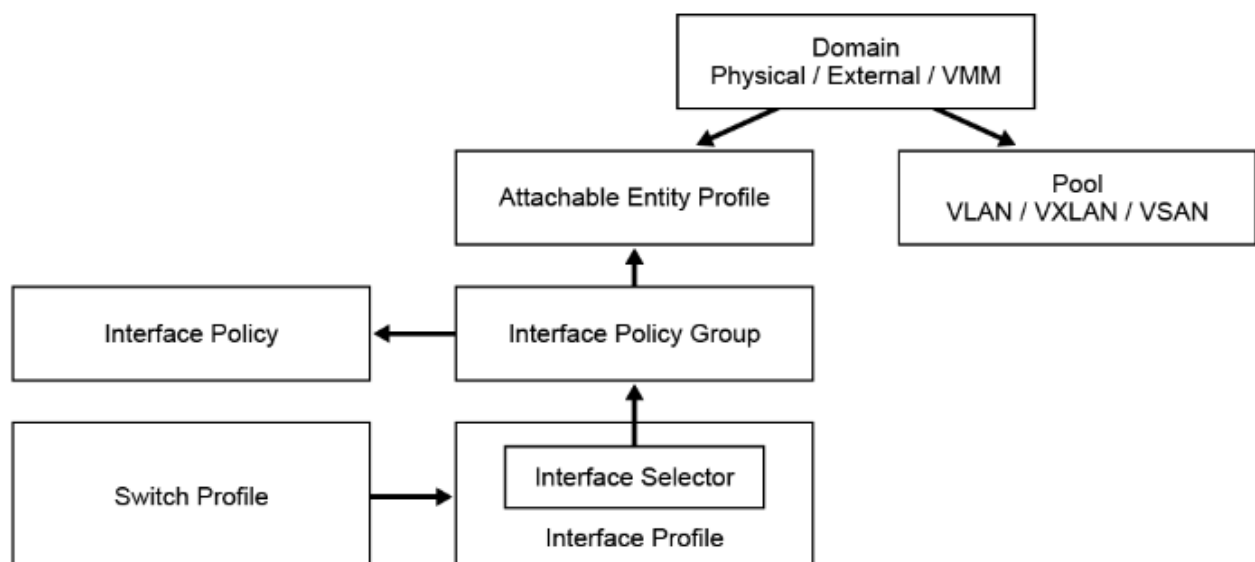


Configure Access Policies for L3Out

The access policies for the external Layer 3 connection include:

- A Layer 3 domain references a VLAN pool, which serves the external Layer 3 connections.
- An attachable access entity profile (AAEP) references a domain, such as the Layer 3 domain, and therefore specifies the VLAN pool that can be activated on an interface.
- An interface policy defines a protocol or interface properties that are applied to interfaces.
- An interface policy group gathers multiple interface policies into one set and binds them to an AAEP.
- The interface selector identifies the interface (or interface block) for the L3Out and associates it with the interface policy group.
- An interface profile groups one or more interface selectors, effectively specifying the policies consumed by the interface blocks.
- A switch profile chooses one or more leaf switches and associates them with an interface profile, effectively specifying the policies consumed by the interface blocks on a given switch.

The following diagram illustrates the relationships between some of these configuration elements:



Go to **Fabric > Access Policies > Pools > VLAN** and create a VLAN pool with the settings below. Click **OK** and **Submit**.

- **Name:** ExtL3_VLANs
- **Allocation mode:** Static Allocation
- Click **+** in the Encap Blocks table and configure the range **51–60** with the default allocation mode (Inherit allocMode from the parent) and role (External or On the wire encapsulations).

The screenshot shows the 'Pools - VLAN' configuration page. A 'Create VLAN Pool' dialog is open, with 'Name' set to 'ExtL3_VLANs', 'Description' set to 'optional', and 'Allocation Mode' set to 'Static Allocation'. Below it, a 'Create Ranges' dialog is open, showing 'Type' as 'VLAN', 'Description' as 'optional', and 'Range' as 'VLAN 51 - VLAN 60'. The 'Allocation Mode' is set to 'Inherit allocMode from parent' and the 'Role' is set to 'External or On the wire encapsulations'. Both dialogs have 'Cancel' and 'OK' buttons.

Click **OK** and **Submit**.

Verify that the VLAN block was created successfully.

The screenshot shows the 'Pools - VLAN' configuration page. The 'VLAN' tab is selected. A table lists the created VLAN pool:

Name	Allocation Mode	Encap Blocks	Description
ExtL3_VLANs	Static Allocation	[51-60]	

Go to **Fabric > Access Policies > Physical and External Domains > L3 Domains**, right-click the menu and choose **Create Layer 3 Domain**.

Create the domain **ExtL3Dom** that references the VLAN pool **ExtL3_VLANs** and click **Submit**.

"A gentleman of engineering must never be satisfied that a thing merely works; he must possess the wisdom to explain why."

Create L3 Domain

Name:

Associated Attachable Entity Profile:

VLAN Pool:

Security Domains:

Select	Name	Description
--------	------	-------------

Go to **Fabric > Access Policies > Policies > Global > Attachable Access Entity Profiles** and create the AAEP **EXTERNAL_SWITCH_AAEP**. Click **+**, add the domain **ExtL3Dom** and click **Update**. Verify the encapsulation range and click **Next** and Finish.

Create Attachable Access Entity Profile

STEP 1 > Profile

Name:

Description:

Enable Infrastructure VLAN: ☐

Domains (VMM, Physical or External) To Be Associated To Interfaces:

Domain Profile	Encapsulation
L3 External Domain - ExtL3Dom	from:vlan-51 to:vlan-60

Go to **Fabric > Access Policies > Interfaces > Leaf Interfaces > Policy Groups > Leaf Access Port**. Create an interface policy group **ExtL3_IPG**, associate it with **EXTERNAL_SWITCH_AAEP** and the Link Layer Discovery Protocol (LLDP) policy **Enable_LLDP**. Click **Submit**.

Create Leaf Access Port Policy Group

Name:

Description:

Attached Entity Profile:

CDP Policy:

Link Level Policy:

LLDP Policy:

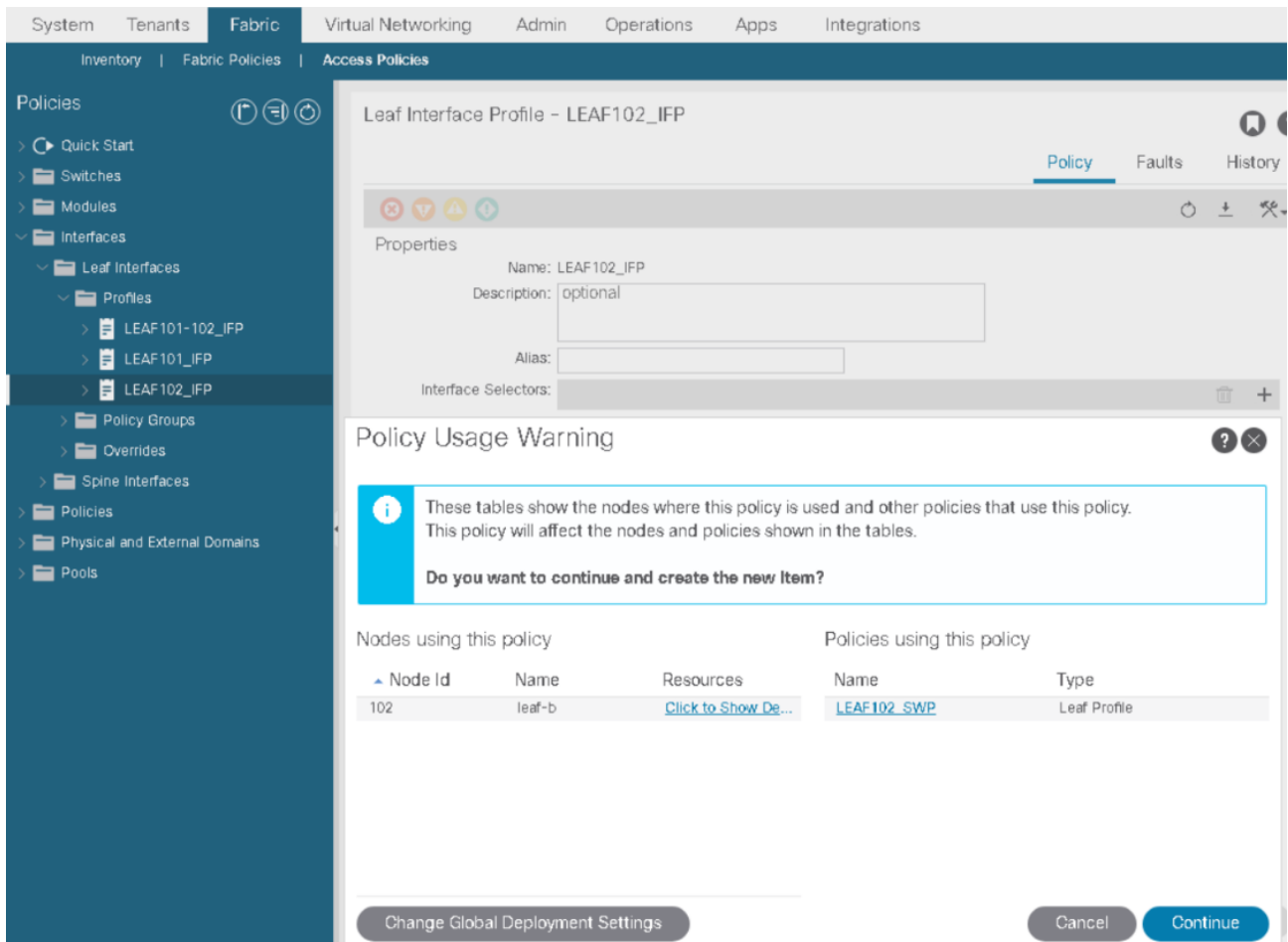
Advanced Settings

802.1x Port Authentication:

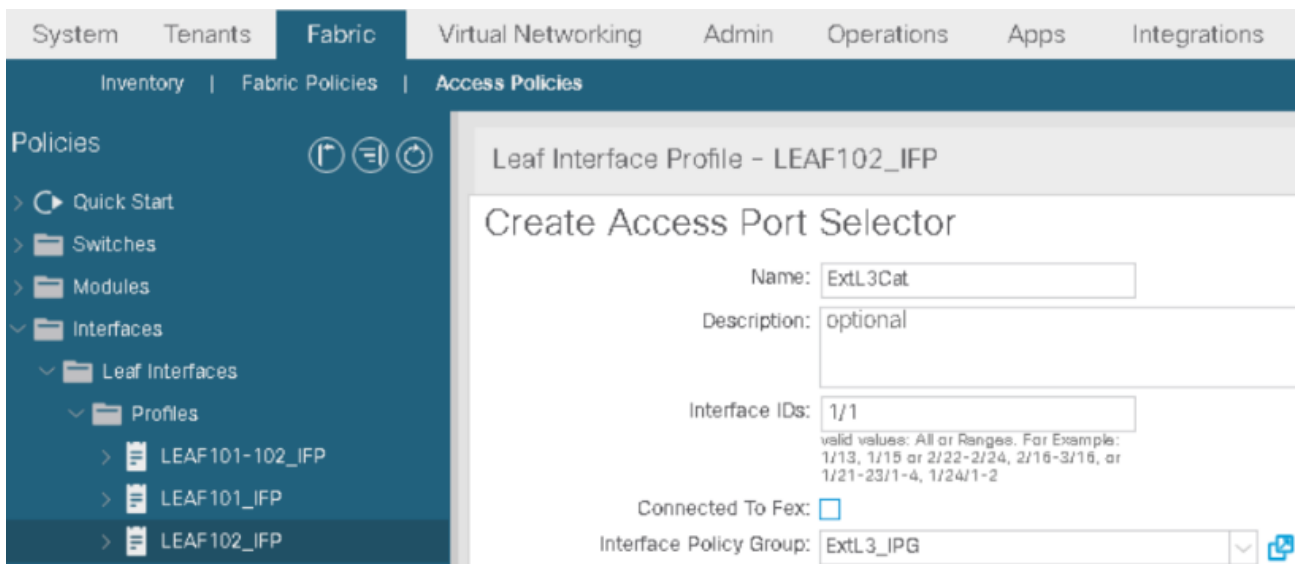
MCP:

Go to **Fabric > Access Policies > Interfaces > Leaf Interfaces > Profiles**, expand **LEAF102_IFP**, click **+** to configure an interface selector and **Continue** in the Policy Usage Warning.

"A gentleman of engineering must never be satisfied that a thing merely works; he must possess the wisdom to explain why."



Configure an interface selector **ExtL3Cat** for the interface ID **1/1** and associate it with the interface policy group **ExtL3_IPG**. Click **Submit**.



Verify LLDP neighborship with the external switch

```
leaf-b# show lldp neig
Capability codes:
  (R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device
  (W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other
Device ID      Local Intf    Hold-time    Capability    Port ID
3560-23        Eth1/1        120          BR            Gi1/0/4
```

"A gentleman of engineering must never be satisfied that a thing merely works; he must possess the wisdom to explain why."

Status will show as "out-of-service".

```
leaf-b# show interface e1/1 status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Eth1/1	--	out-of-ser	trunk	full	1G	1000base-T

```
leaf-b#
```

```
leaf-b# show interface e1/1
```

```
Ethernet1/1 is up (out-of-service)
admin state is up, Dedicated Interface
```

Note: the out-of-service status is expected, and it will be resolved once the interface has been attached to the L3Out configuration.

Login the external switch to verify the configurations in place.

```
3560-23>show ip ospf interface brief
```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Lo0	1	0	172.16.100.100/32	1	LOOP	0/0	
Vl51	1	0	172.16.1.2/30	1	DR	0/0	

```
3560-23>
```

```
3560-23>show lldp neig | in leaf-b
```

leaf-b	Gil/0/4	120	B,R	Eth1/1
--------	---------	-----	-----	--------

```
3560-23>
```

```
3560-23>show interface trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gil/0/3	on	802.1q	trunking	1
Gil/0/4	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Gil/0/3	21,31
Gil/0/4	51

Port	Vlans allowed and active in management domain
Gil/0/3	21,31
Gil/0/4	51

Port	Vlans in spanning tree forwarding state and not pruned
Gil/0/3	21,31
Gil/0/4	51

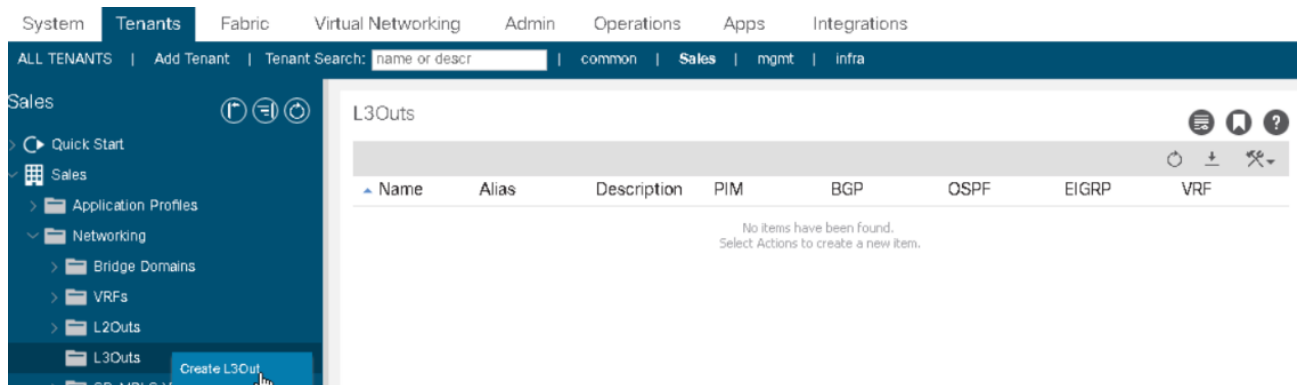
Configure L3Out

Having configured the required access policies under the Layer 3 domain for L3Out, you can configure OSPF on the interface and will enable an OSPF exchange between leaf-b and the external Cisco Catalyst switch.

The border leaf supports various area types, such as a not-so-stubby area (NSSA) or regular areas, and additional features, such as authentication. The external switch has been preconfigured for the backbone area without authentication.

The OSPF adjacency will be established through the configuration of a L3Out.

Go to **Tenants > Sales > Networking > L3Outs**, right-click the menu, and choose **Create L3Out**.



Start creating a L3Out with the settings below. Click **Next**.

- **Name:** OSPF_L3Out
- **VRF:** Presales_VRF
- **Layer 3 Domain:** ExtL3Dom
- **Routing protocol:** OSPF
- **OSPF Area ID:** 0 (backbone)
- **OSPF Area Type:** Regular area

Create L3Out

1. Identity | 2. Nodes And Interfaces | 3. Protocols | 4. External

Leaf --- Route

Identity

A Layer 3 Outside (L3Out) network configuration defines how the ACI fabric connects to external layer 3 networks. The L3Out supports connecting to external networks using static routing and dynamic routing protocols (BGP, OSPF, and EIGRP).

Prerequisites:

- Configure an L3 Domain and Fabric Access Policies for interfaces used in the L3Out (AAEP, VLAN pool, Interface selectors).
- Configure a BGP Route Reflector Policy for the fabric infra MP-BGP.

Name:

VRF:

L3 Domain:

Use for GOLP: ☐

☐ BGP ☐ EIGRP ☒ OSPF

OSPF Area ID:

OSPF Area Control: ☒ Send redistributed LSAs into NSSA area
☒ Originate summary LSA
☐ Suppress forwarding address in translated LSA

OSPF Area Type:

OSPF Area Cost:

Enter these settings in the *Nodes and Interfaces* page and then click **Next**.

- Clear the Use Defaults checkbox. This will allow you to enter a custom node profile name.
- **Node Profile Name:** L102

- **Layer 3: SVI.** Recall the interface types for external Layer 3 connections: routed interfaces, routed subinterfaces, switched virtual interfaces (SVIs), and floating SVIs. This lab will use the SVI option, which allow multiple connections over a single physical link.
- **Layer 2: Port**
- **Node ID:** leaf-b (Node-102)
- **Router ID:** 10.1.1.1
- Delete the Loopback Address. You do not need to configure a loopback on the border leaf. Only the router ID is mandatory. In case of BGP, if the BGP peering needs to be sourced from a loopback, you may use this option.
- **Interface:** eth1/1
- **Interface Profile Name:** *OSPF_L3Out_interfaceProfile*. The interface profile is a container for interface settings. You can keep the default name.
- **Encap:** VLAN
- **VLAN ID:** 51. 802.1Q tagging supports multiple logical connections over the physical link. The subinterface with the tag 51 is used for L3Out connectivity. This VLAN ID must belong to the static VLAN pool associated with the Layer 3 domain (51-60) that was assigned to the L3Out.
- **MTU:**1500 (the maximum value supported on the Catalyst switch). The default value on the Cisco ACI switches is 9216. The MTU mismatch would prevent the OSPF adjacency from being established.
- **IP Address:**172.16.1.1/30. It will be applied as the IP address on the switch virtual interface (SVI) on the border leaf.

Create L3Out

1. Identity 2. Nodes And Interfaces 3. Protocols 4. External EPG

Nodes and Interfaces

The L3Out configuration consists of node profiles and interface profiles. An L3Out can span across multiple nodes in the fabric. All nodes used by the L3Out can be included in a single node profile and is required for nodes that are part of a VPC pair. Interface profiles can include multiple interfaces. When configuring dual stack interfaces a separate interface profile is required for the IPv4 and IPv6 configuration, that is automatically taken care of by this wizard.

Use Defaults: ☒

Interface Types

Layer 3: Interface Sub-Interface **SVI** Floating SVI

Layer 2: **Port** Virtual Port Channel Direct Port Channel

Nodes

Node ID leaf-b (Node-102)	Router ID 10.1.1.1	Loopback Address 	+ Hide Interfaces
Leave empty to not configure any Loopback.			
Interface eth1/1 Ex: eth1/1 or topology/pod-1/paths-101/peerip-leaf1/23	Encap VLAN	VLAN ID 51 Integer Value	MTU (bytes) 1500
IP Address 172.16.1.1/30 address/mask		+	

In the *Protocols* page, click **Next** without choosing any protocol policy.

Create L3Out

1. Identity

2. Nodes And Interfaces

3. Protocols

Protocol Associations

OSPF

Node ID: 102

Hide Policy ☐

Interface

1/1

Policy:

select a value

In the *External EPG* page, configure an external EPG with these settings and click **Finish**:

- **Name:** *Cat_ExtNet*. An external network instance profile (External EPG, L3Out EPG) represents a group of external subnets that have the same security behavior.
- **Default EPG for all external networks:** *checked*. This is a placeholder for all networks (0.0.0.0/0), to apply the consumed contract to any external network through an external EPG.

Create L3Out

1. Identity

2. Nodes And Interfaces

3. Protocols

4. External EPG

External EPG

The L3Out Network or External EPG is used for traffic classification, contract associations, and route control policies. Classification is matching external networks to this EPG for applying contracts. Route control policies are used for filtering dynamic routes exchanged between the ACI fabric and external devices, and leaked into other VRFs in the fabric.

Name: Cat_Ext

Provided Contract: select a value

Consumed Contract: select a value

Default EPG for all external networks: ☒

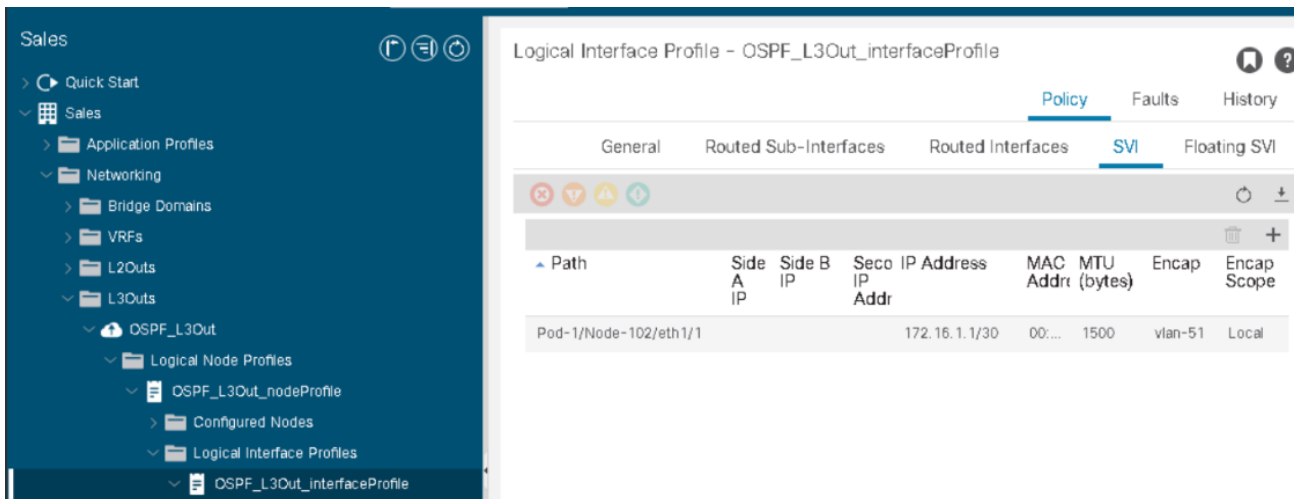
Click Finish

Expand the L3Out and examine its sub-components, the logical node profile, and logical interface profile.

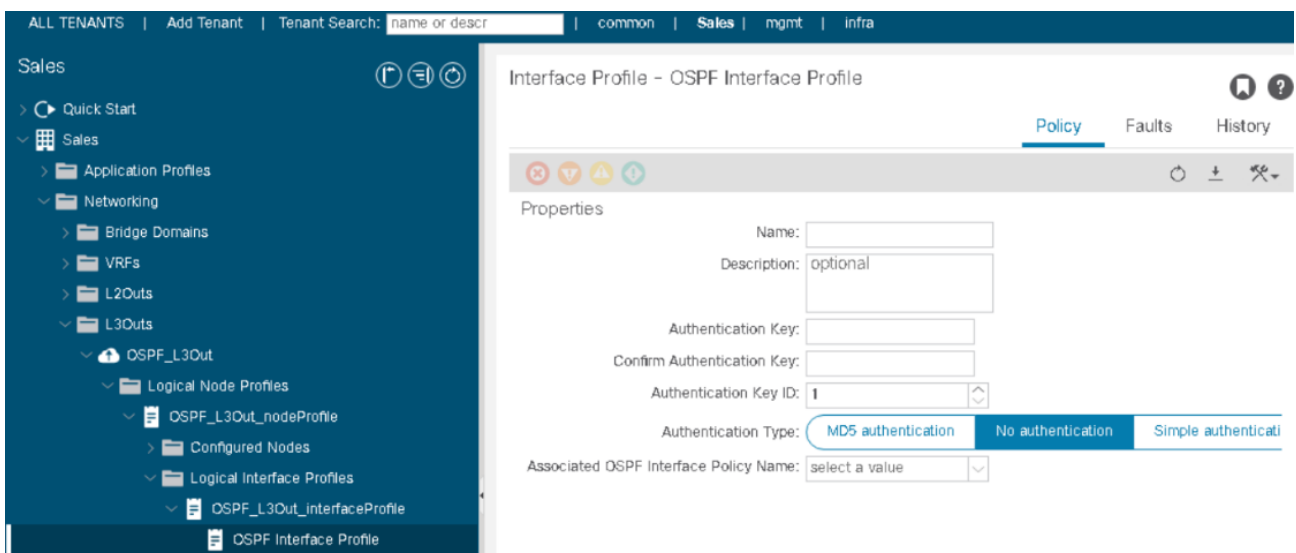
The screenshot shows the Cisco ACI GUI with the 'Tenants' tab selected. The left sidebar shows the navigation tree with 'Sales' expanded, and 'Logical Node Profiles' under 'OSPF_L3Out' selected. The main panel displays the 'Logical Node Profile - OSPF_L3Out_nodeProfile' configuration page. The 'Policy' tab is active, showing the 'Properties' section. The 'Name' is 'OSPF_L3Out_nodeProfile', 'Description' is 'optional', 'Alias' is empty, and 'Target DSCP' is 'Unspecified'. The 'Nodes' section shows a table with two columns: 'Node ID' and 'Router ID'. One node is listed: 'topology/pod-1/node-102' with 'Router ID' '10.1.1.1'.

Node ID	Router ID
topology/pod-1/node-102	10.1.1.1

"A gentleman of engineering must never be satisfied that a thing merely works; he must possess the wisdom to explain why."

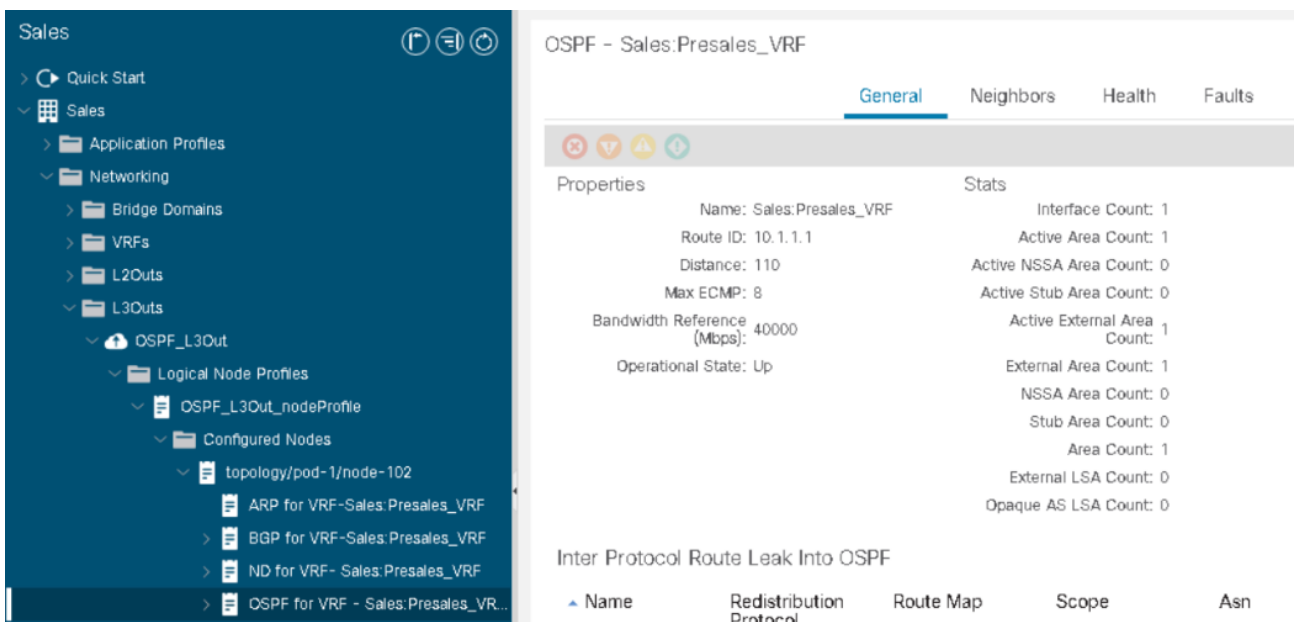


Expand the logical interface profile and verify the OSPF interface profile configuration.



The OSPF interface profile is equivalent to the **ip router ospf area 0** command in Cisco Nexus Operating System (NX-OS). Without it, OSPF would be disabled on that interface.

Below the logical node profile, expand the **Configured Nodes** menu



"A gentleman of engineering must never be satisfied that a thing merely works; he must possess the wisdom to explain why."

OSPF - Sales:Presales_VRF			
Healthy			
Neighbor Id	State	Peer Ip	Interface
172.16.100.100	Full	172.16.1.2	vlan9

Expand the **External EPGs**, choose **Cat_ExtNet** and examine the subnet information in the **Policy** > **General** tab.

Sales

Quick Start

Sales

Application Profiles

Networking

Bridge Domains

VRFs

L2Outs

L3Outs

OSPF_L3Out

Logical Node Profiles

OSPF_L3Out_nodeProfile

External EPGs

Cat_Ext

Route map for import and exp...

SR-MPLS VRF L3Outs

Dot1Q Tunnels

Contracts

Policies

Services

Security (Beta)

External EPG - Cat_Ext

Policy

Operational

General

Contracts

Inherited Contracts

Properties

Description: optional

pcTag: 32771

Contract Exception Tag:

Configured VRF Name: Presales_VRF

Resolved VRF: uni/tn-Sales/ctx-Presales_VRF

QoS Class: Unspecified

Target DSCP: Unspecified

Configuration Status: applied

Configuration Issues:

Preferred Group Member:

Exclude

Include

Intra Ext-EPG Isolation:

Enforced

Unenforced

Subnets:

IP Address

Scope

0.0.0.0/0

External Subnets for the External EPG

The action "**0.0.0.0/0 with External Subnets for the External EPG scope**" causes all endpoints to be classified into the same EPG. You could define more external EPGs if you wanted to apply a different policy for different groups of external endpoints. In this case, all outside endpoints are treated equally and one external EPG network for all subnets (0.0.0.0/0) was created.

Leaf102 interface status (now in connected state) (no longer out-of-ser as before).

```
leaf-b# show interface e1/1 status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Eth1/1	--	connected	trunk	full	1G	1000base-T

"A gentleman of engineering must never be satisfied that a thing merely works; he must possess the wisdom to explain why."

On the Catalyst switch, verify the OSPF adjacency resulting from your L3Out configuration.

```
3560-23>show ip ospf neig
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.1.1.1	1	FULL/BDR	00:00:30	172.16.1.1	Vlan51

```
3560-23>
```

On leaf-b View the IP OSPF interfaces in your VRF.

```
leaf-b# show ip ospf neig vrf Sales:Presales_VRF
```

```
OSPF Process ID default VRF Sales:Presales_VRF
Total number of neighbors: 1
Neighbor ID      Pri  State           Up Time  Address        Interface
172.16.100.100   1   FULL/DR         00:14:41  172.16.1.2     Vlan9
leaf-b#
```

```
leaf-b# show ip ospf interface vrf Sales:Presales_VRF
```

```
Vlan9 is up, line protocol is up
  IP address 172.16.1.1/30, Process ID default VRF Sales:Presales_VRF, area backbone
  Enabled by interface configuration
  State BDR, Network type BROADCAST, cost 4
  Index 77, Transmit delay 1 sec, Router Priority 1
  Designated Router ID: 172.16.100.100, address: 172.16.1.2
  Backup Designated Router ID: 10.1.1.1, address: 172.16.1.1
  1 Neighbors, flooding to 1, adjacent with 1
  Timer intervals: Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello timer due in 00:00:09
  No authentication
  Number of opaque link LSAs: 0, checksum sum 0
leaf-b#
```

You can verify the OSPF operation in the Cisco Application Policy Infrastructure Controller (APIC) user interface.

Go to **Fabric > Inventory > Pod ID**, expand the node (leaf-b) and choose **Protocols > OSPF**.

The screenshot displays the Cisco APIC web interface. The left sidebar shows the navigation tree with 'Fabric' selected, and 'Inventory' expanded to show 'leaf-b (Node-102)'. Under 'Protocols', 'OSPF' is selected, and 'OSPF for VRF - Sales:Presales_VRF (default)' is expanded. The main panel shows the 'OSPF - Sales:Presales_VRF' configuration page with tabs for 'General', 'Neighbors', 'Health', and 'Faults'. The 'General' tab is active, showing properties like Name, Route ID, Distance, Max ECMP, Bandwidth Reference, and Operational State. A 'Stats' section on the right lists various OSPF statistics. Below the stats, the 'Inter Protocol Route Leak Into OSPF' table is visible, showing route leak configurations for different protocols.

Name	Redistribution Protocol	Route Map	Scope	Asn
Sales:Presales_VRF	BGP	exp-cbx-proto-23592...	Inter protocol leak	65001
Sales:Presales_VRF	COOP	exp-cbx-st-2359296	Inter protocol leak	1
Sales:Presales_VRF	Direct	exp-cbx-st-2359296	Inter protocol leak	1
Sales:Presales_VRF	EIGRP	exp-cbx-proto-23592...	Inter protocol leak	1
Sales:Presales_VRF	Static	exp-cbx-st-2359296	Inter protocol leak	1

"A gentleman of engineering must never be satisfied that a thing merely works; he must possess the wisdom to explain why."

On **leaf-b**, view the OSPF routes.

```
leaf-b# show ip route ospf vrf Sales:Presales_VRF
IP Route Table for VRF "Sales:Presales_VRF"
'*' denotes best ucast next-hop
'**' denotes best mcast next-hop
'[x/y]' denotes [preference/metric]
'%<string>' in via output denotes VRF <string>

172.16.100.100/32, ubest/mbest: 1/0
    *via 172.16.1.2, vlan9, [110/5], 00:16:30, ospf-default, intra
```

The network 172.16.100.100/32 has been received over the L3Out from the Catalyst switch. The connecting interface (vlan9 in this example) shows the Platform Independent (PI) VLAN ID.

```
leaf-b# show vlan extended
```

VLAN	Name	Encap	Ports
1	Sales:eCommerce_AP:App_EPG	vlan-12	Eth1/3, Po4
8	infra:default	vxlان-16777209, vlan-3967	Eth1/2
9	Sales:Presales_VRF:l3out- OSPF_L3Out:vlan-51	vxlان-15007704, vlan-51	Eth1/1

The PI VLAN is associated with a Virtual Extensible LAN (VXLAN) ID and the encap VLAN (51) that is used on the Eth1/1 trunk.

Configure Spine to Act as MP-BGP Route Reflector

Within the Cisco ACI fabric, Multiprotocol Border Gateway Protocol (MP-BGP) is implemented between leaf and spine switches to propagate external routes within the Cisco ACI fabric. The BGP route reflector technology is deployed to support many leaf switches within a single fabric. All the leaf and spine switches are in a single BGP AS. When a border leaf learns about external routes, it redistributes the external routes of a given VRF to an MP-BGP address family. MP-BGP maintains a separate BGP routing table for each VRF. Within MP-BGP, the border leaf advertises the routes to the spine switch that acts as a BGP route reflector. The routes are then propagated to all the leaves where the VRFs are instantiated.

Here a BGP route reflector policy is created by specifying the BGP AS number and the spine node that should act as the BGP route reflector. Cisco APIC will then automatically enable Internal Border Gateway Protocol (IBGP) peering between leaves and spine (or spines) and configure leaf switches as route reflector clients. Cisco APIC will also automatically generate the required configuration for route redistribution on the border leaves.

In the Cisco APIC user interface, go to **System > System Settings > BGP Route Reflector**.

- Set the autonomous system to 65001.
- Add the spine ID (**201**) to the Route Reflector Nodes and click **Submit**, **Submit**, and **Submit Changes**.

System | Tenants | Fabric | Virtual Networking | Admin | Operations | Apps | Integrations

QuickStart | Dashboard | Controllers | **System Settings** | Smart Licensing | Faults | Config Zones | Events | Audit Log | Active Sessions

System Settings

- APIC Connectivity Preferences
- APIC Passphrase
- BD Enforced Exception List
- BGP Route Reflector**
- Control Plane MTU
- COOP Group
- Date and Time
- Endpoint Controls
- Fabric Security
- Fabric-Wide Settings
- Global AES Passphrase Encryption Set...
- Global Endpoints (Beta)
- Intersight
- ISIS Policy
- Load Balancer
- Port Tracking

BGP Route Reflector Policy - BGP Route Reflector

Policy | Faults

Properties

Name: default
Description: optional

Autonomous System Number: 65001

Domain ID Base: 0

Route Reflector Nodes:

Pod ID	Node ID	Node Name	Description
1	201	spine	

Create Route Reflector Node

Spine Node: spine

Description: spine
Pod - 1/201

Verify:

BGP Route Reflector Policy - BGP Route Reflector

Policy

Properties

Name: default
Description: optional

Autonomous System Number: 65001

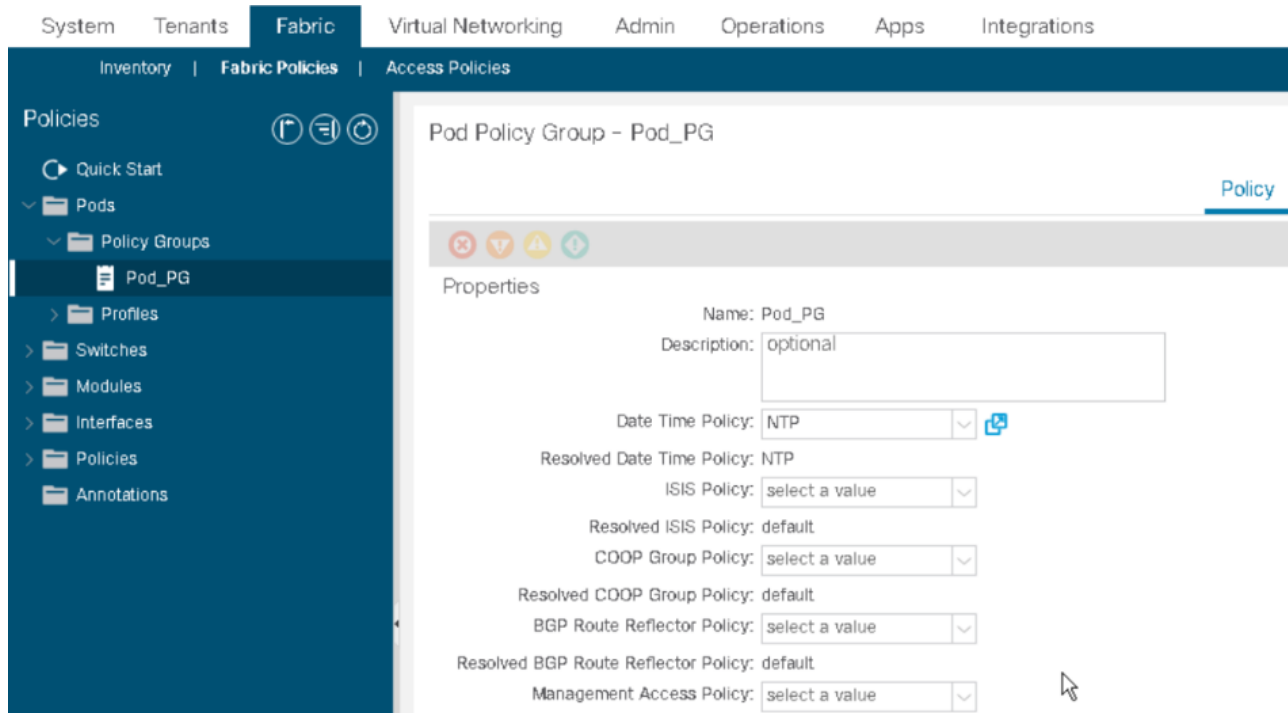
Domain ID Base: 0

Route Reflector Nodes:

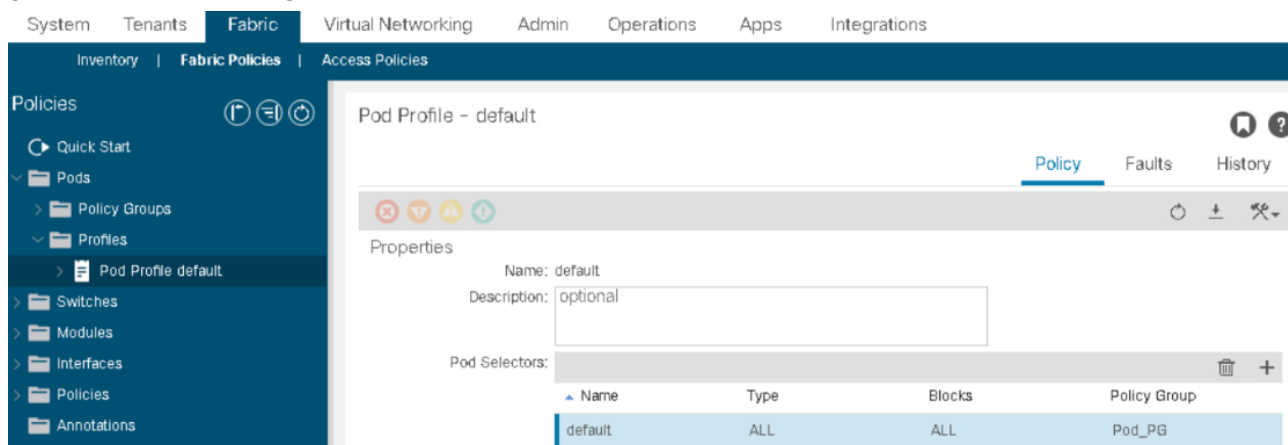
Pod ID	Node ID	Node Name	Description
1	201	spine	

Go to **Fabric > Fabric Policies > Pods > Policy Groups**.

Verify that the policy group **Pod_PG** resolves the BGP Route Reflector Policy to **default**.



Go to **Fabric > Fabric Policies > Pods > Profiles > Pod Profile default** and verify that the policy group **Pod_PG** is assigned to the pod selector **default**.



On any leaf switch, verify the MP-BGP sessions.

```
leaf-b# show bgp sessions vrf overlay-1
Total peers 1, established peers 1
ASN 65001
VRF overlay-1, local ASN 65001
peers 1, established peers 1, local router-id 10.0.216.66
State: I-Idle, A-Active, O-Open, E-Established, C-Closing, S-Shutdown

Neighbor      ASN      Flaps  LastUpDn|LastRead|LastWrit  St  Port (L/R)  Notif (S/R)
10.0.216.65   65001    0      00:00:35|never   |never   E   42831/179   0/0
leaf-b#
```

You can verify BGP sessions in the Cisco APIC user interface, in **Fabric > Inventory > Pod 1 > spine > Protocols > BGP > BGP for VRF-overlay-1 > Sessions**.

System Tenants **Fabric** Virtual Networking Admin Operations Apps Integrations

Inventory Fabric Policies Access Policies

Inventory

- Quick Start
- Topology
- Pod 1
 - leaf-a (Node-101)
 - leaf-b (Node-102)
 - spine (Node-201)
 - Chassis
 - Interfaces
 - Control Plane Statistics
 - Protocols
 - BGP
 - BGP for VRF-manage...
 - BGP for VRF-overlay-1
 - Sessions
 - BGP NextHops
 - VPNv4
 - Neighbors

Sessions

Addr	State	Peer Asn
10.0.216.64	Established	65001
10.0.216.66	Established	65001

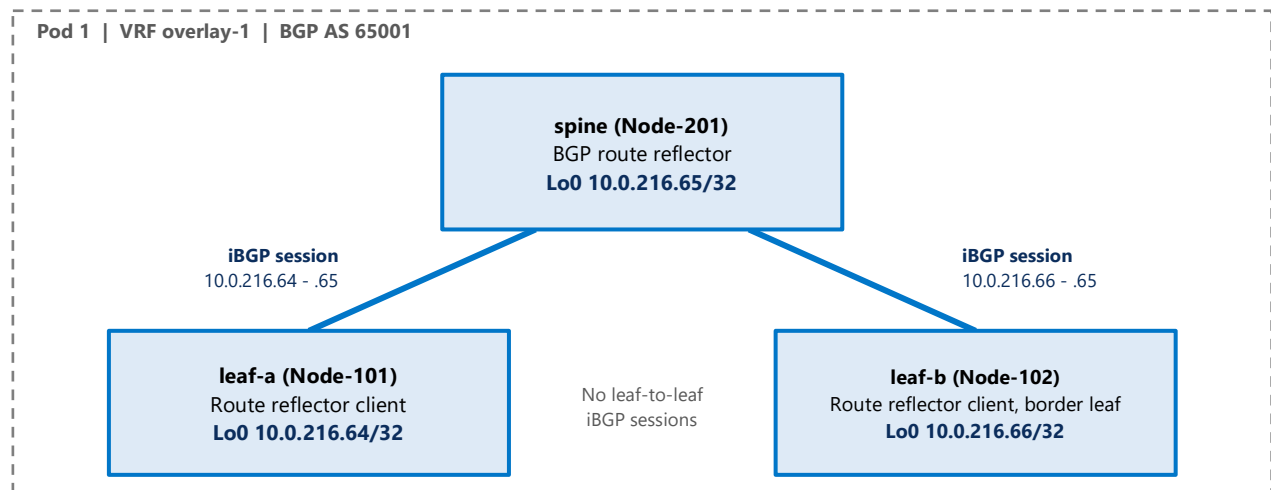


Figure: iBGP peerings in AS 65001. Spine-201 is the route reflector, and leaf-a and leaf-b peer only with it. The addresses are the lo0 loopbacks of each node.

On **leaf-b**, verify route redistribution from OSPF to MP-BGP.

```

leaf-b# show bgp ipv4 unicast vrf Sales:Presales_VRF
BGP routing table information for VRF Sales:Presales_VRF, address family IPv4 Unicast
BGP table version is 8, local router ID is 10.0.1.254
Status: s-suppressed, x-deleted, S-stale, d-dampened, h-history, *-valid, >-best
Path type: i-internal, e-external, c-confed, l-local, a-aggregate, r-redist, I-injected
Origin codes: i - IGP, e - EGP, ? - incomplete, | - multipath, & - backup

  Network          Next Hop        Metric      LocPrf     Weight Path
*>r172.16.1.0/30    0.0.0.0          0           100       32768 ?
*>r172.16.100.100/32 0.0.0.0          5           100       32768 ?
  
```

The route 172.16.100.100/32 was received via OSPF from the Catalyst switch, while 172.16.1.0/30 is the directly connected SVI subnet. ACI redistributes both into MP-BGP.

Verify MP-BGP route propagation on leaf-a.

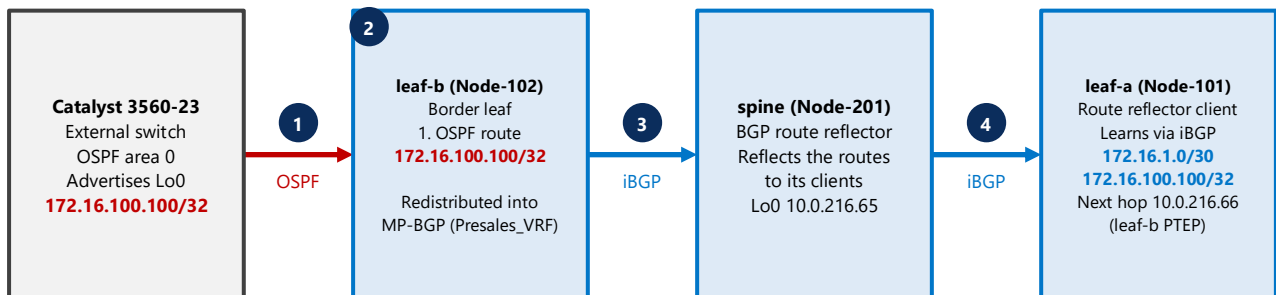
"A gentleman of engineering must never be satisfied that a thing merely works; he must possess the wisdom to explain why."

```
leaf-a# show bgp ipv4 uni vrf Sales:Presales_VRF
BGP routing table information for VRF Sales:Presales_VRF, address family IPv4 Unicast
BGP table version is 4, local router ID is 10.0.1.254
Status: s-suppressed, x-deleted, S-stale, d-dampened, h-history, *-valid, >-best
Path type: i-internal, e-external, c-confed, l-local, a-aggregate, r-redist, I-injected
Origin codes: i - IGP, e - EGP, ? - incomplete, | - multipath, & - backup

  Network          Next Hop          Metric      LocPrf      Weight Path
*>i172.16.1.0/30    10.0.216.66        0           100         0 ?
*>i172.16.100.100/32 10.0.216.66        5           100         0 ?
```

The BGP next-hop IP address is the leaf-b PTEP address.

Route flow: external OSPF route into ACI MP-BGP and on to leaf-a



- 1 The Catalyst advertises its loopback 172.16.100.100/32 to leaf-b in OSPF area 0.
- 2 leaf-b redistributes the OSPF route and the connected SVI subnet 172.16.1.0/30 into MP-BGP in Presales_VRF.
- 3 leaf-b sends both routes over iBGP to spine-201, the route reflector.
- 4 The spine reflects them to leaf-a, which installs them with next hop 10.0.216.66 (the leaf-b PTEP).

Figure: Route flow from the Catalyst to leaf-a. The OSPF route is redistributed into MP-BGP on leaf-b, passed to the spine route reflector over iBGP and reflected to leaf-a.

BGP will be automatically enabled on any leaf that has an external Layer 3 network attached, and also on any leaf where the VRF associated with the Layer 3 external network is instantiated (leaves that do not have the VRF associated preserve the hardware resources by not running BGP).

On **leaf-a**, verify that L3Out VXLAN does not exist here.

```
leaf-a# show vlan extended
```

VLAN	Name	Encap	Ports
8	infra:default	vxlan-16777209, vlan-3967	Eth1/2
17	Sales:Presales_BD	vxlan-16351138	Eth1/1, Eth1/3, Po4
19	Sales:DB_BD	vxlan-15925207	Eth1/3, Po4
20	Sales:eCommerce_AP:Web_EPG	vlan-21	Eth1/1
21	Sales:eCommerce_AP:Web_EPG	vlan-11	Eth1/3, Po4
22	Sales:eCommerce_AP:DB_EPG	vlan-13	Eth1/3, Po4
23	Sales:eCommerce_AP:App_EPG	vlan-12	Eth1/3, Po4

Advertise BD Subnet and Enable EPG and Ext-EPG Communication

To enable bidirectional communication between a BD endpoint and the external network, you will associate the L3Out with the bridge domain DB_BD and advertise the BD subnet externally. You will create a new subnet (10.0.3.254/24) for this purpose.

Go to **Tenants > Sales > Networking > Bridge Domains > DB_BD > Policy > L3 Configurations**. In the Subnets table, click + and create a new subnet with the settings below. Click Submit.

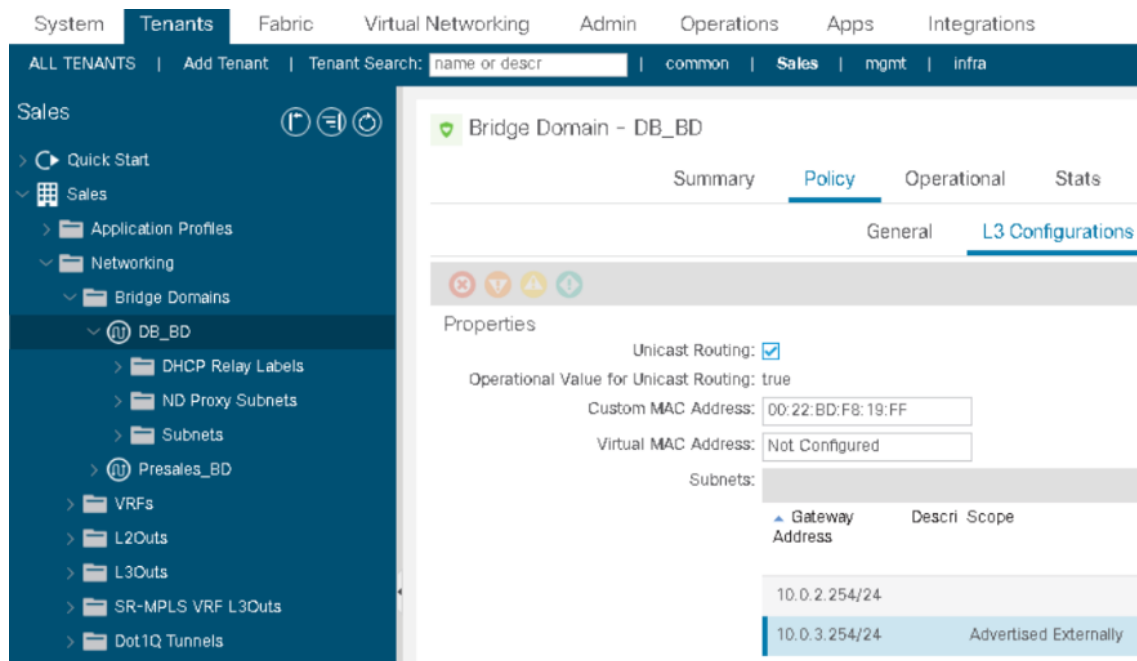
"A gentleman of engineering must never be satisfied that a thing merely works; he must possess the wisdom to explain why."

Gateway IP: 10.0.3.254/24

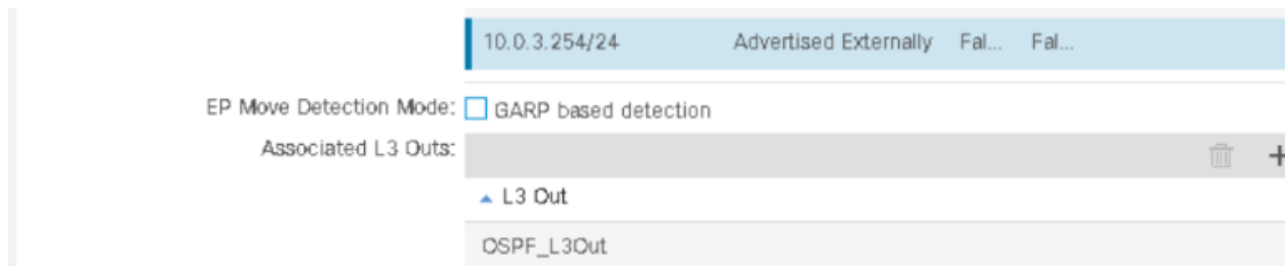
Scope: Advertised Externally

The Advertised Externally scope allows the border leaf to advertise this subnet out of the associated L3Out. With the default scope, Private to VRF, the subnet would remain inside the fabric, and the Catalyst switch would not learn it.

Verify that the subnet 10.0.3.254/24 is listed in the Subnets table with the scope Advertised Externally.



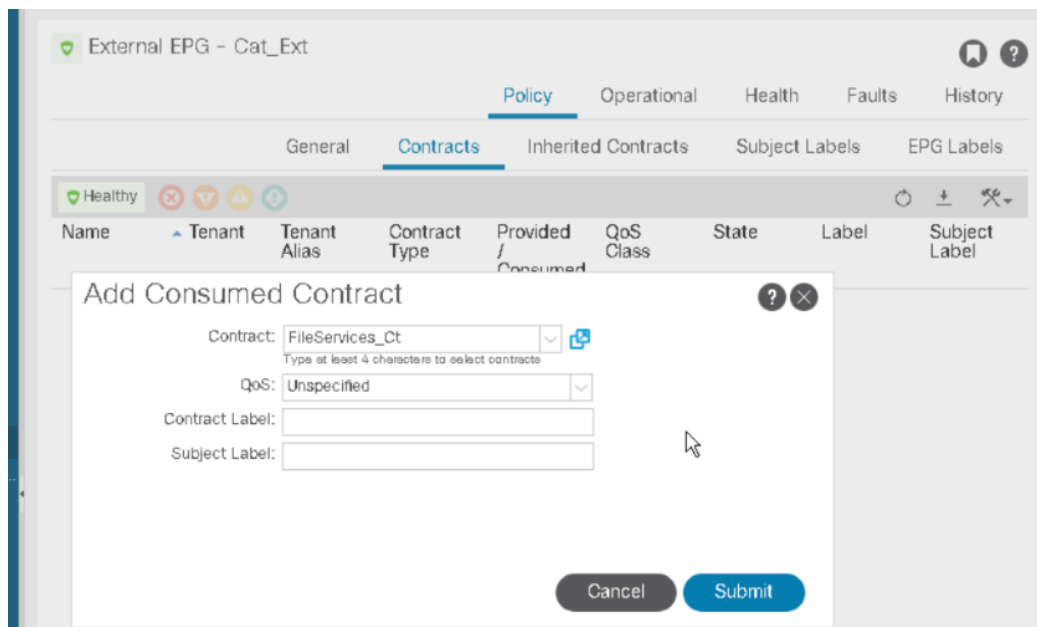
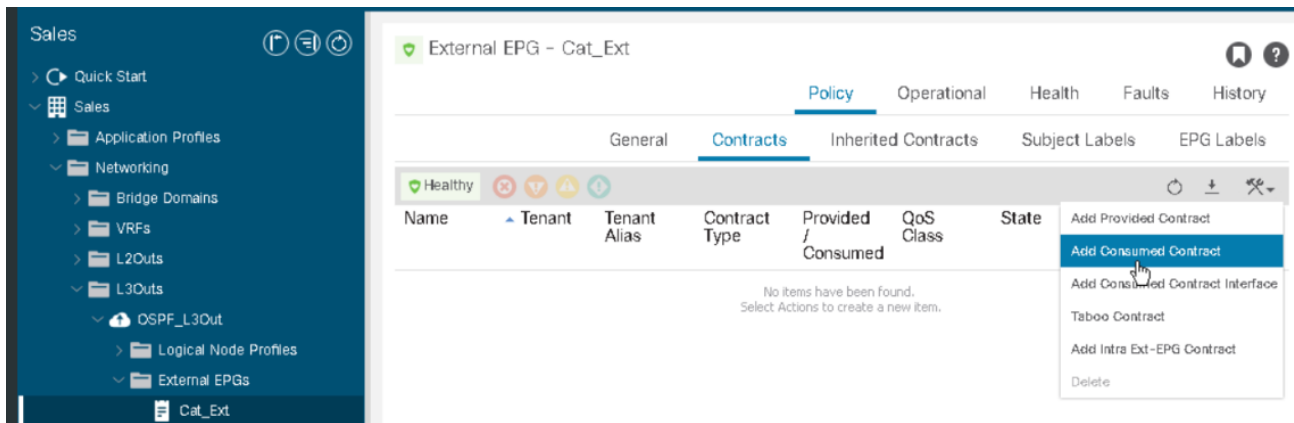
Click the plus sign (+) in the Associated Layer 3 Outs table, choose **OSPF_L3Out**, and click **Update**.



Then you will apply the FileServices_Ct contract, which permits ICMP, SSH and FTP, as a consumer contract to the external network EPG. The contract is already applied as a provider contract to DB_EPG.

Note: The implementation of the FileServices_Ct contract (its subjects and filters) is not in the scope of this document. The contract is assumed to already exist.

Go to **Networking > L3Outs > OSPF_L3Out > External EPGs > Cat_ExtNet** and choose **Policy > Contracts**. Click the tools button, choose **Add Consumed Contract**, choose **FileServices_Ct** and then click **Submit**.



On the Catalyst switch, verify the OSPF routing table.

```
3560-23>show ip route ospf
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O E2 10.0.3.0/24 [110/20] via 172.16.1.1, 00:06:27, Vlan51
```

To verify reachability through the L3Out, ping the internal host 10.0.3.1, which belongs to the 10.0.3.0/24 subnet, from the Catalyst switch.

```
3560-23>ping 10.0.3.1
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 10.0.3.1, timeout is 2 seconds:
```

```
.....
```

```
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/2/4 ms
```

```
3560-23>
```

"A gentleman of engineering must never be satisfied that a thing merely works; he must possess the wisdom to explain why."

References

1. <https://www.cisco.com/site/us/en/learn/training-certifications/training/courses/dcaci.html>
2. <https://www.cisco.com/c/en/us/solutions/collateral/data-center-virtualization/application-centric-infrastructure/guide-c07-743150.html>
3. <https://titusm.github.io/Cisco-Data-Center/aci/l3out-bgp-ospf/>